## Digital Communications

## Digital Transmission

In *Digital transmission*, the information (either analog or digital originally) is transmitted digitally.

**Advantages**

- Noise immunity
- Ease of processing and multiplexing.
- Simpler to measure and evaluate.
- The use of signal regenerators rather than signal amplifiers.

**Disadvantages**

- Greater bandwidth required.
- The need for ADC (transmitter side) and DAC(receiver side) in case original information is analog.
- Precise timing or synchronization is strictly required.



## Pulse Modulation

*Pulse modulation* is used in digital transmission. Pulse modulation is divided into two fields: Analog Pulse Modulation and Digital Pulse Modulation. Examples of analog pulse modulation are:

- Pulse Width Modulation (PWM) or Pulse Length Modulation (PLM) or Pulse Duration Modulation(PDM)
- Pulse Position Modulation (PPM)
- Pulse Amplitude Modulation (PAM)
- Pulse Frequency Modulation (PFM)

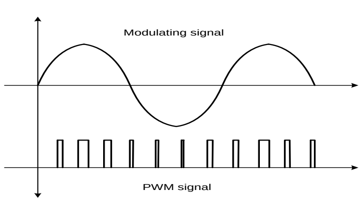
For digital pulse modulation, the two predominant methods (with some of their variations) are:

- Pulse Code Modulation (PCM)
- Differential PCM (DPCM)
- Delta Modulation (DM)
- Adaptive Delta Modulation (ADM)



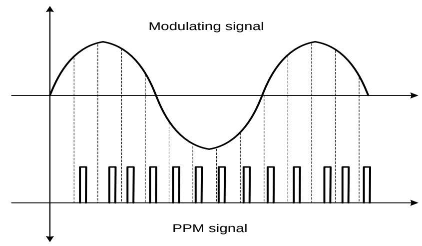
## Pulse Width Modulation (PWM)

In PWM, the pulse width (active portion of the duty cycle) is proportional to the amplitude of the analog modulating signal. The recent designation of PWM is L3E (formerly P3E).




## Pulse Position Modulation (PPM)

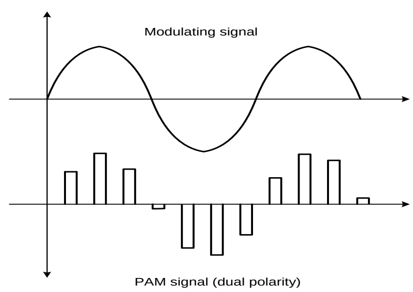
In PPM, the position of a constant-width pulse within a prescribed time slot is varied in accordance with the amplitude of the analog modulating signal. The recent designation of PPM is M3E (formerly P3F).




## Pulse Amplitude Modulation (PAM)

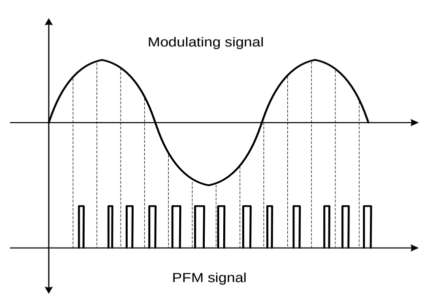
In PAM, the amplitude of a constant-width, constant-position pulse is varied in accordance with the amplitude of the analog modulating signal. PAM is an intermediate form of modulation used in either PSK, QAM and PCM systems. Its recent designation is K3E (formerly P3D).






## Pulse Frequency Modulation (PFM)

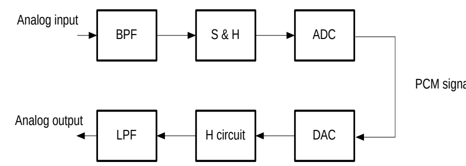
Pulse frequency modulation is a pulse modulation system in which the amplitude of the pulse is kept constant while its period and duration are made proportional to the modulating signal so that the duty cycle of the pulse train remains constant. Its designation is V3E.




## Pulse Code Modulation (PPM)

In *PCM*, the analog signal is sampled and converted to pulses of constant length and encoded for serial binary transmission. The binary number varies in accordance with the amplitude of the analog modulating signal. Its designation is *P3G*.

### The PCM System




## The Nyquist Sampling Theorem

The *Nyquist sampling theorem* establishes the minimum sampling rate of PCM systems. It states that for an analog signal to be recovered and reproduced accurately at the receiver, it must be sampled at least twice its highest frequency. This is the minimum sampling rate.



$$f_s = 2f_a$$

where:

$f_s$ = Nyquist Sample Rate

$f_a$ = highest frequency to be sampled

If the sampling rate $f_s$ is less than $f_a$, distortion results. The distortion is called ***aliasing*** *or* ***foldover distortion***.

In practice, the actual sampling rate is made higher or greater than the minimum $f_s$.

$$f_s > 2f_a$$

## Dynamic Range

The *dynamic range* is the ratio of the largest possible magnitude to the smallest possible magnitude that can de decoded by the digital to analog converter.

$$DR = \frac{V_{max}}{V_{min}} = \frac{V_{max}}{\text{resolution}}$$

$$DR(dB) = 20\log DR$$

where:



$V_{max}$ = largest possible magnitude that can be decoded

$V_{min}$ = smallest possible magnitude that can be decoded

### PCM Code Number of Bits

The *PCM code* is sign-magnitude code where the most significant bit (MSB) is the sign bit and the remaining bits are magnitude bits. It is sometimes called a *folded binary code* except for the sign bit.